

Background and Objective

- Any degree of deformation or asymmetry has an estimated incidence as high as 50%¹
- Heterogenous presentation: **~85% are unilateral¹**
- Traditionally treated via reconstructive surgery
- Can be molded within the first 2-3 months²



Market Analysis

$$\begin{matrix} \text{Prevalence of correctable} \\ \text{malformations} \sim \mathbf{1:6000^3} \end{matrix} \times \begin{matrix} \sim \mathbf{15 \text{ million}} \text{ births/year} \\ \text{in developed markets} \end{matrix} \times \begin{matrix} \sim \mathbf{4,000 \text{ USD}} \text{ per} \\ \text{AuriForm series} \end{matrix} = \mathbf{10 \text{ Million} \text{ USD/Year}}$$

Assuming ~30%
capture rate
= **3 million USD**
yearly with
minimal hardware
requirements or
provider training

Solutions Landscape

EarWell®

- Expensive
- Limited Customization
- Good modeling capability
- Uncomfortable



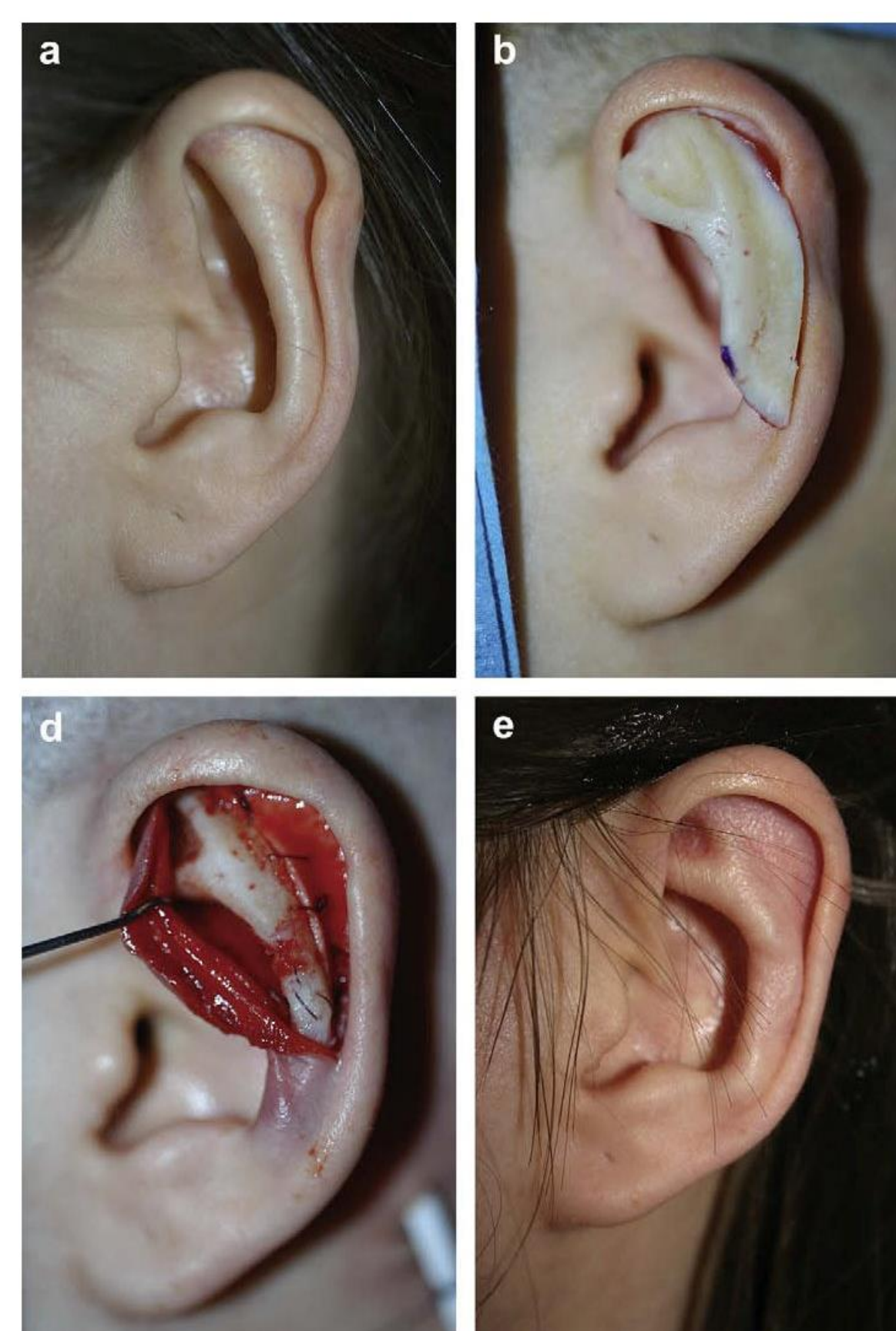
Wire and Tape

- Inexpensive
- Customizable
- Experience dependent
- Limited modeling capability



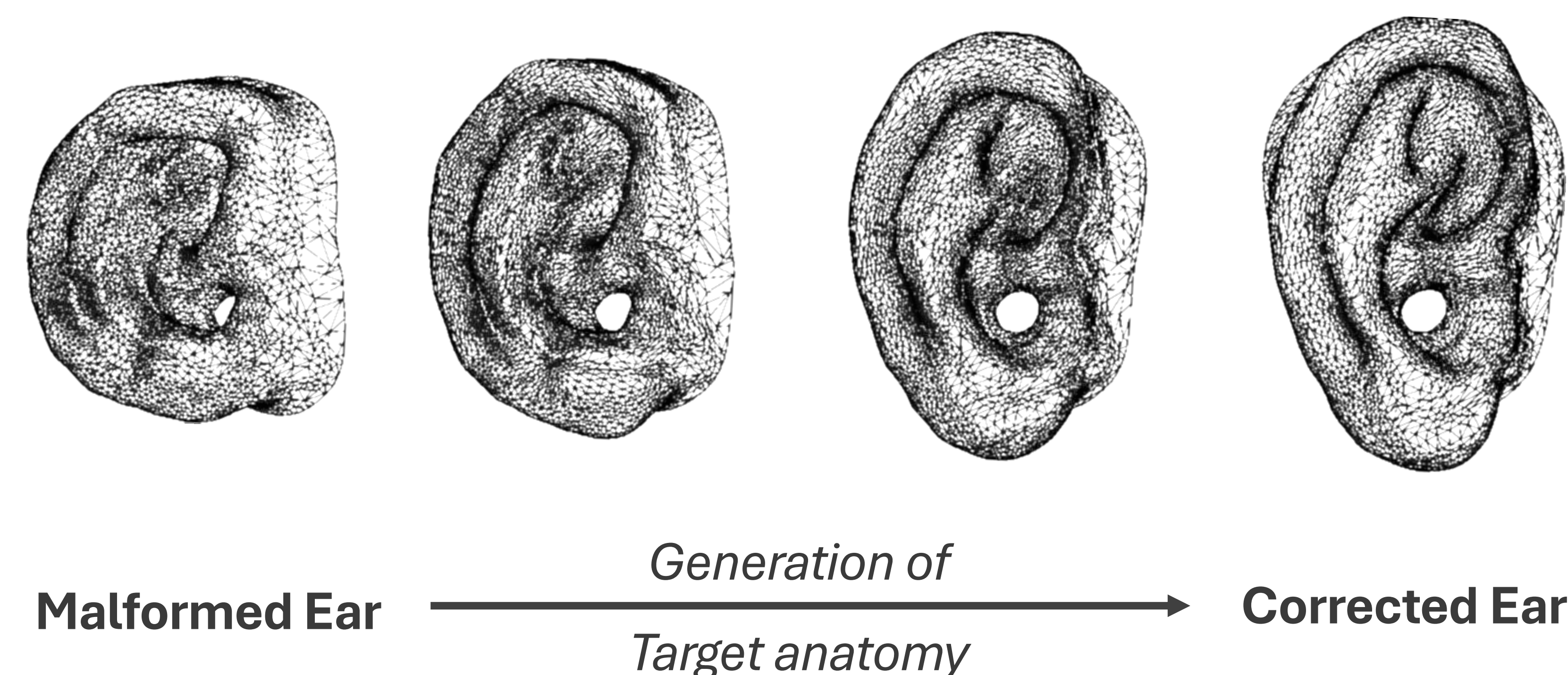
Otoplasty

- Gold standard
- Can fix complex defects
- Most expensive
- Age agnostic



Prototyping and Software

Step-wise target generation



Scanning, molds, and fit testing

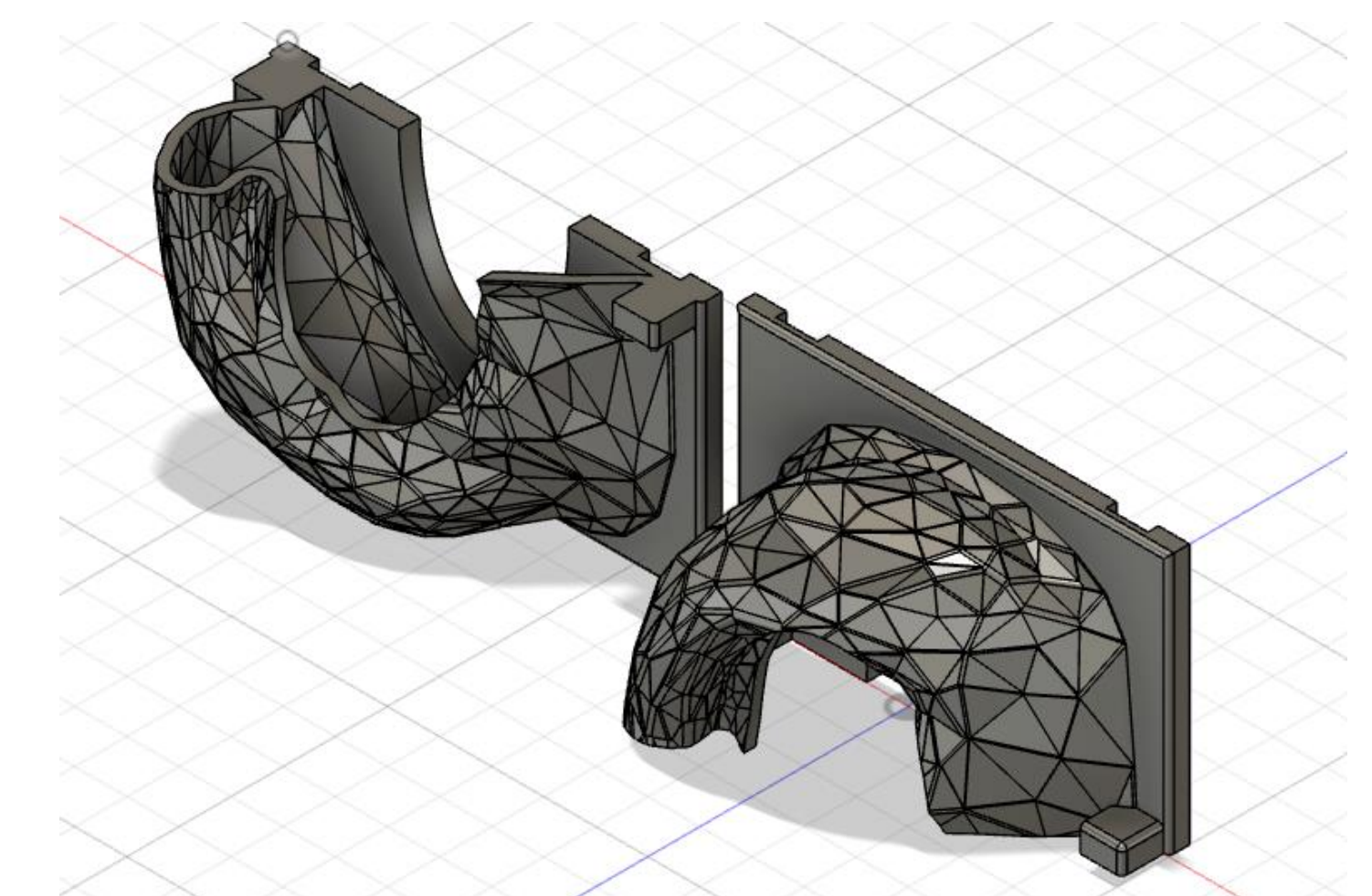


Silicone Fit Testing



Next Steps

- Algorithm:** Refine generation of intermediate anatomy and fully automate mold generation with tweakable parameters
- Hardware:** Acquire more sophisticated LIDAR scanners
- Data:** Capture wide range of congenital malformations

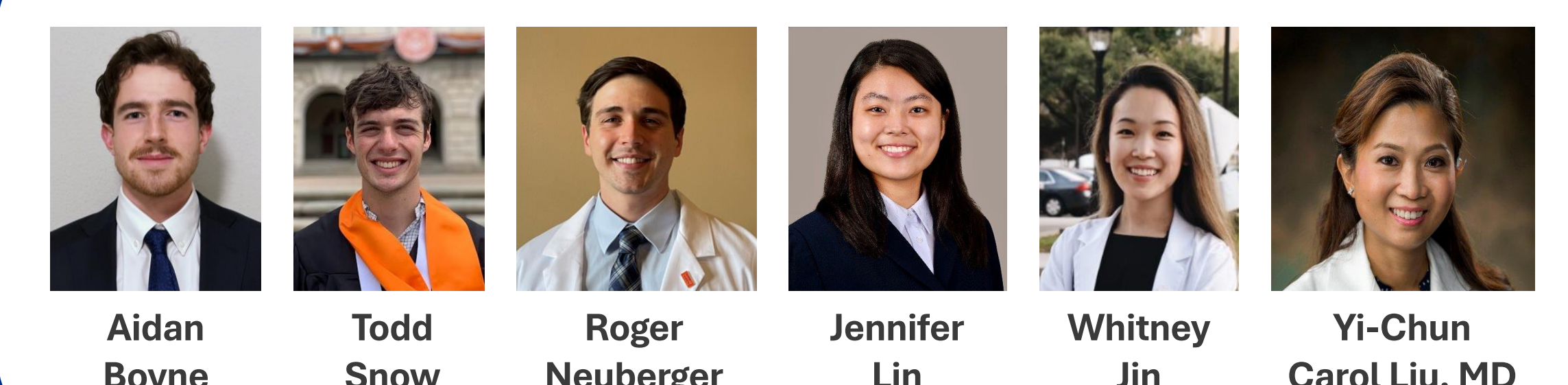


- Materials:** Identify more durable and skin-friendly polymers for the 3D-printed molds
- In-Vivo:** Conduct trials on living tissue to establish efficacy and expected treatment course

References

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- Bartel-Friedrich S, Wulke C. Classification and diagnosis of ear malformations. *GMS Curr Top Otorhinolaryngol Head Neck Surg*. 2007;6:Doc05. Epub 2008 Mar 14. PMID: 22073081; PMCID: PMC3199848.

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